Lecture 11 – Heat and Thermal Stress

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In this lecture we consider what thermometers really measure and the condition for thermal equilibrium, and how the dimensions of an object change as a result of a temperature change. In particular, we will evaluate the thermal stress on materials due to such expansions or contraction caused by heat.

The slides in this lecture come from Chapter-17 and Chapter-11 of Young's textbook.

This PPT file will well define the scope of your tests.

1

Introduction

- Does this molten iron at 1500 degrees Celsius contain heat?
- The terms "temperature" and "heat" have very different meanings, even though most people use them interchangeably.
- Heat is energy.
- Temperature is a value on a scale.

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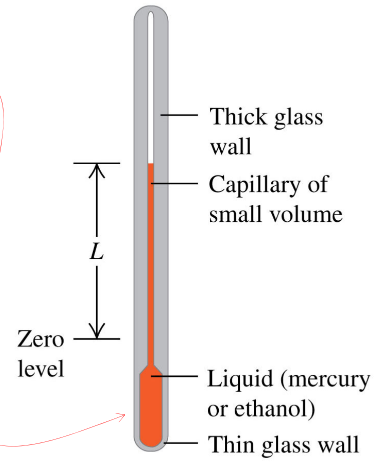
Temperature and Thermal Equilibrium

- We use a **thermometer** to measure **temperature**.
- For example, the volume of the liquid in the thermometer corresponds to the right changes with temperature.
- Two systems are in **thermal equilibrium** if and only if they have the same temperature.



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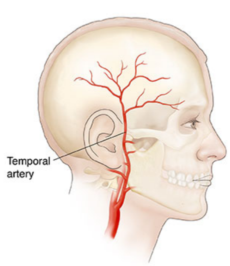
Other Types of Thermometers

- A temporal artery thermometer measures infrared radiation from the skin that overlies one of the important arteries in the head.
- Although the thermometer's cover touches the skin, the infrared detector inside the cover does not.



This is wrong position!!

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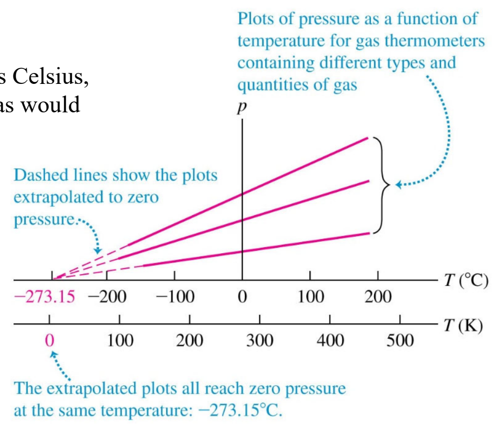
Temperature Scales

- On the **Celsius** (or **centigrade**) **temperature scale**, 0 degrees Celsius is the freezing point of pure water, and 100 degrees Celsius is its boiling point.

Absolute Zero

There is a temperature, −273.15 degrees Celsius, at which the absolute pressure of any gas would become zero.



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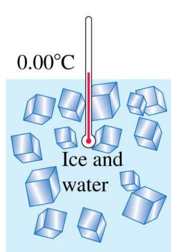
Temperature Scales

- On the **Kelvin** (or **absolute**) **temperature scale**, 0 K is the extrapolated temperature at which a gas would exert no pressure.
- To convert from Celsius to Kelvin:

Kelvin temperature → $T_K = T_C + 273.15$ ← Celsius temperature

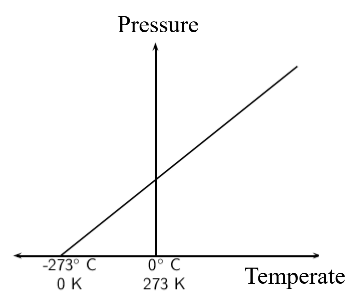


Kelvin temperatures are measured in kelvins ...

$T = 273.15$ K ◀ **RIGHT!**

... *not* "degrees" kelvin.

$T = 273.15$ ~~°K~~ ◀ **WRONG**



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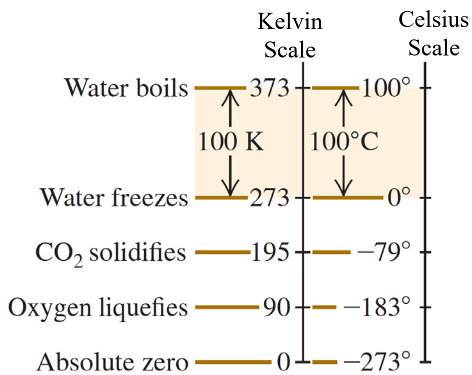
Temperature Conversions

- Below are relationships between the Kelvin (K) and Celsius (C) temperature scales. Temperatures have been rounded off to the nearest degree.



Kelvin temperature

$$T_K = T_C + 273.15$$

Celsius temperature

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Linear Thermal Expansion

- Increasing the temperature of a rod causes it to expand.
- For moderate changes in temperature, the change in length is given by:

Linear thermal expansion:

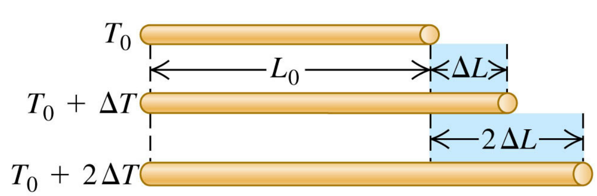
Change in length

$$\Delta L = \alpha L_0 \Delta T$$

Original length

Temperature change

Coefficient of linear expansion



This formulation is due to the observation in the lab experiment.

Literal Expansion

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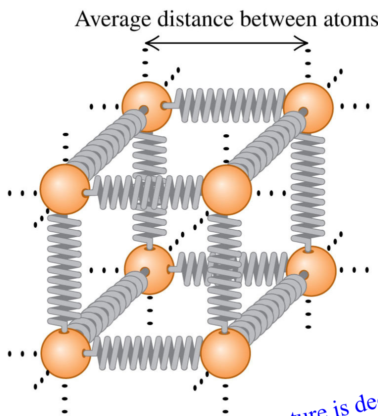
Molecular Basis for Thermal Expansion

- We can understand linear expansion if we model the atoms as being held together by springs.
- When the temperature increases, the average distance between atoms also increases.
- As the atoms get farther apart, every dimension increases.

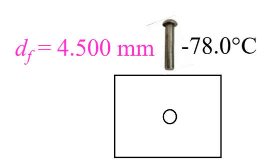


What if the temperature is decreased?
We will then expect a contraction.

Problem. Aluminum rivets used in airplane construction are made slightly larger than the rivet holes cooled by dry ice (solid CO_2) before being driven. If the diameter of a hole is 4.500 mm, what should be the diameter of a rivet at 23.0°C if its diameter is to equal that of the hole when the rivet is cool to -78.0 ° C (the temperature of dry ice)? $\left(\alpha_{\text{Aluminum}} = 2.4\times10^{-5}\,(°\text{C})^{-1}\right)$

Answer:

Let d_i be the diameter at 23.0°C, and d_f be the diameter at –78.0°C.



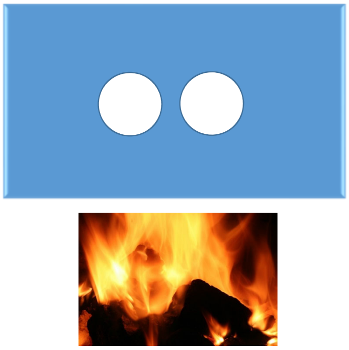
$$\Delta T = T_f - T_i = (-78°\text{C}) - 32°\text{C} = -110°\text{C}$$

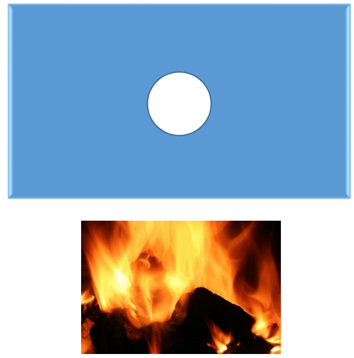
$$d_f = d_i + \Delta d = d_i + \alpha d_i \Delta T = d_i\left(1 + \alpha\Delta T\right)$$

$$4.5\times10^{-3} = d_i(1 + 2.4\times10^{-5}\times(-110))$$

$$0.0045 = d_i \times 0.99736 \rightarrow d_i = 0.004511\ \text{m} = 4.511\ \text{mm}$$

Heating Two Holes in a Plate





Will the holes become bigger or smaller?

Expanding Holes and Volume Expansion

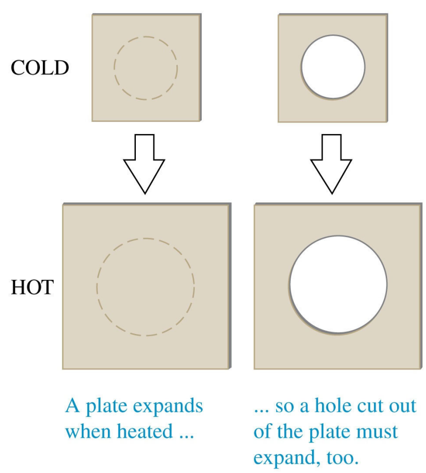
Effect of heat

- If an object has a hole in it, the hole also expands with the object as shown.
- The hole does **not shrink**.

- The change in volume due to thermal expansion is given by

$$\Delta V = \beta V_0 \Delta T$$

where β is the **coefficient of volume expansion** and is equal to 3α.



Coefficient of Linear Expansion (α) versus Coefficient of Volume Expansion (β)

Let L be the length. Let the volume $V = L^3$. Let ΔT be the change in temperature, and the corresponding change in volume is ΔV. Let $\Delta V = \beta V_i(\Delta T)$.

$$V_f = V_i + \Delta V = (L_i + \Delta L)^3 = L_i^3 + 3L_i^2(\Delta L) + 3L_i\cancel{(\Delta L)^2} + \cancel{(\Delta L)^3}$$

First Principle Approach

$$\rightarrow V_f = L_i^3 + 3L_i^2(\alpha L_i(\Delta T))$$

$$= L_i^3 + 3L_i^3(\alpha(\Delta T))$$

$$= V_i + 3\alpha V_i(\Delta T)$$

$$\rightarrow \Delta V = 3\alpha V_i(\Delta T) \quad \rightarrow \quad \beta = 3\alpha$$

coefficient of volume expansion

coefficient of linear expansion

Or from $V = L^3$, $\dfrac{dV}{dL} = 3L^2$ $\rightarrow$ $\Delta V = 3L_i^2(\Delta L)$

$$= 3L_i^2(\alpha L_i(\Delta t)) = 3\alpha L_i^3(\Delta t)$$

$$= 3\alpha V_i(\Delta t) \rightarrow \beta = 3\alpha$$

Calculus Approach

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Coefficients of Linear Expansion

Material	α[K^{-1} or ($^\circ$C)$^{-1}$]
Aluminum	2.4×10^{-5}
Brass	2.0×10^{-5}
Copper	1.7×10^{-5}
Glass	$0.4 - 0.9 \times 10^{-5}$
Invar (nickel–iron alloy)	0.09×10^{-5}
Quartz (fused)	0.04×10^{-5}
Steel	1.2×10^{-5}

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Coefficients of Volume Expansion

Solids	$\beta[K^{-1}$ or $(^{\circ}C)^{-1}]$
Aluminum	7.2×10^{-5}
Brass	6.0×10^{-5}
Copper	5.1×10^{-5}
Glass	$1.2-2.7\times10^{-5}$
Invar	0.27×10^{-5}
Quartz (fused)	0.12×10^{-5}
Steel	3.6×10^{-5}

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Coefficients of Linear Expansion

Coefficient of Volume Expansion

Material	$\alpha[K^{-1}$ or $(^{\circ}C)^{-1}]$	$\beta[K^{-1}$ or $(^{\circ}C)^{-1}]$
Aluminum	2.4×10^{-5}	7.2×10^{-5}
Brass	2.0×10^{-5}	6.0×10^{-5}
Copper	1.7×10^{-5}	5.1×10^{-5}
Glass	$0.4-0.9\times10^{-5}$	$1.2-2.7\times10^{-5}$
Invar (nickel–iron alloy)	0.09×10^{-5}	0.27×10^{-5}
Quartz (fused)	0.04×10^{-5}	0.12×10^{-5}
Steel	1.2×10^{-5}	3.6×10^{-5}

X 3

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Problem. A geodesic dome constructed with an aluminum framework is a nearly perfect hemisphere; its diameter measures 54.0 m on a winter day at a temperature of -12° C. How much more interior space does the dome have in the summer where the temperature is 35° C?

$\left(\beta_{\text{Aluminum}} = 7.2\times10^{-5}\,(°\text{C})^{-1}\right)$

Answer:

Let V_i be the volume of the hemisphere at $-12°\text{C}$, $V_i = \dfrac{2}{3}\pi r_i^3$.

Let V_f be the volume at 35.0°C.

$$\Delta T = T_f - T_i = 35°\text{C} - (-12°\text{C}) = 47°\text{C}$$

$$\Delta V = \beta V_i \Delta T = 7.2\times10^{-5}\times\left(\frac{2}{3}\pi\left(\frac{54}{2}\right)^3\right)\times 47 = 139.50\text{ m}^3$$

Or, $r_f = r_i\left(1+\alpha\Delta T\right) = 27\left(1+\left(2.4\times10^{-5}\right)\times 47\right) = 27.030456\text{ m}$

$$\Delta V = V_f - V_i = \frac{2}{3}\pi\left(27.030456^3 - 27^3\right) = 139.66\text{ m}^3$$

Slight deviation due to the large magnitude of ΔT, thus the effect due to the omission of $(\Delta T)^2$ and $(\Delta T)^3$ are accumulated.

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Example of Thermal Expansion

- This railroad track has a gap between segments to allow for thermal expansion.
- On hot days, the segments expand and fill in the gap.
- If there were no gaps, the track could buckle under very hot conditions.

MRT Track at Red Hill Station

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Thermal Expansion

Thermal Stress $\left(\frac{\text{N}}{\text{m}^2}\right)$

- If we clamp the ends of a rod rigidly to prevent expansion or contraction, the change in temperature will result in **thermal stress** in the rod.

- Expansion joints on bridges are needed to accommodate changes in length that result from thermal expansion.

Thermal stress:

Force needed to keep length of rod constant

$$\frac{F}{A} = -Y\alpha\,\Delta T$$

Young's modulus

Temperature change

Coefficient of linear expansion

Cross-sectional area of rod

What is young's modulus?

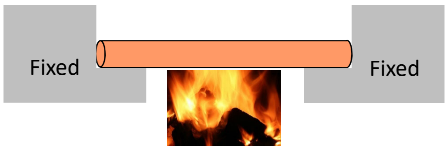
Young's modulus is a measure of the ability of a material to withstand changes in length when under lengthwise tension or compression.

Thermal Stress $\left(\frac{\text{N}}{\text{m}^2}\right)$

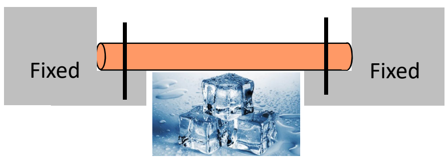
Thermal stress:

$$\underbrace{\frac{F}{A}}_{\substack{\text{Force needed to keep length of rod constant} \\ \text{Cross-sectional area of rod}}} = -\underbrace{Y}_{\text{Young's modulus}}\,\underbrace{\alpha}_{\text{Coefficient of linear expansion}}\,\underbrace{\Delta T}_{\text{Temperature change}}$$

- The thermal stress in an object is due to the external constraint that inhibits the elongation or the contraction of the object.

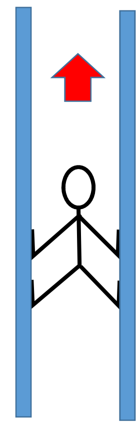


If the long cylinder is being heated in a fixed space, it will experience thermal stress (compressive stress)



If the long cylinder is being cooled with its ends pinned into a fixed space, it will experience thermal stress (tensile stress)

compressive stress or tensile stress on this spiderman ?



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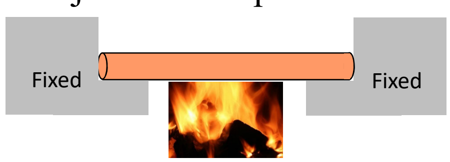
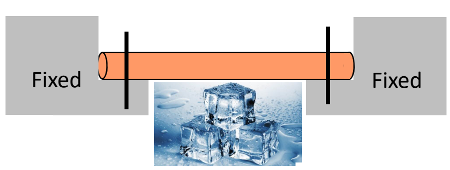
The compressive stress in materials is due to the constrained space thus inhibiting the object from expansion due to the increase in temperature.



The tensile stress in materials is due to the constrained movement, such as being pinned at both ends of the object thus inhibiting the object from contraction due to the decrease in temperature.



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Stress and Strain

- When you pinch your nose, the force per area that you apply to your nose is called **stress**.
- The fractional (ratio) change in the size of your nose is called **strain**.
- The deformation is **elastic** because your nose springs back to its initial size when you stop pinching.

What is young's modulus?

Tensile Stress and Strain

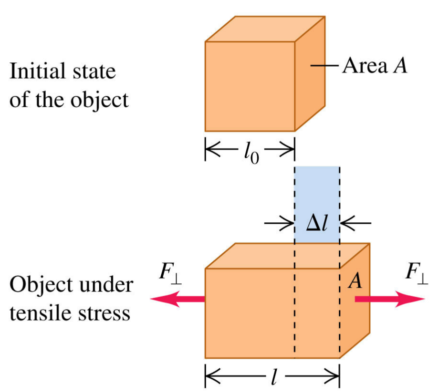
If the object is of length l_0 and cross sectional area A, and is being pulled in opposite directions by external agents at both ends with forces of equal magnitude $F_\perp$, the tensile stress on the object is defined as:

$$\text{tensile stress} = \frac{F_\perp}{A}$$

and the object is in tension.

If the elongation is Δl, the tensile strain on the object is defined as

$$\text{tensile strain} = \frac{\Delta l}{l_0}$$



$$\text{Tensile stress} = \frac{F_\perp}{A}$$

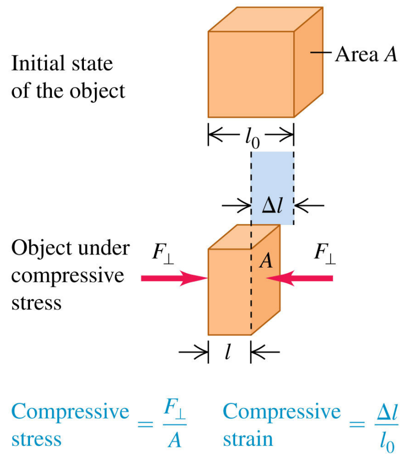
What is young's modulus?

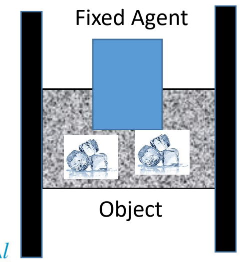
Tensile Stress and Strain

If the external forces in the diagram are reversed, the object will be pushed rather than pulled. The object will be under compression and the compressive stress and compressive strain are defined in the same way, except that Δl is the distance that object contracts.

$$\text{compressive stress} = \frac{F_\perp}{A}$$

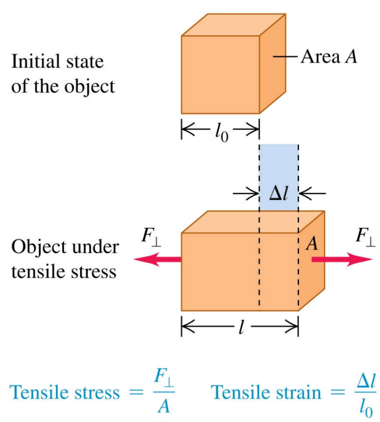
$$\text{compressive strain} = \frac{\Delta l}{l_0}$$





Young's Modulus

- Experiment shows that for a sufficiently small tensile stress, stress and strain are proportional.
- The corresponding elastic modulus is called **Young's modulus**.



Young's modulus for tension

Force applied perpendicular to cross section

Original length

$$Y = \frac{\text{Tensile stress}}{\text{Tensile strain}} = \frac{F_\perp / A}{\Delta l / l_0} = \frac{F_\perp}{A}\frac{l_0}{\Delta l}$$

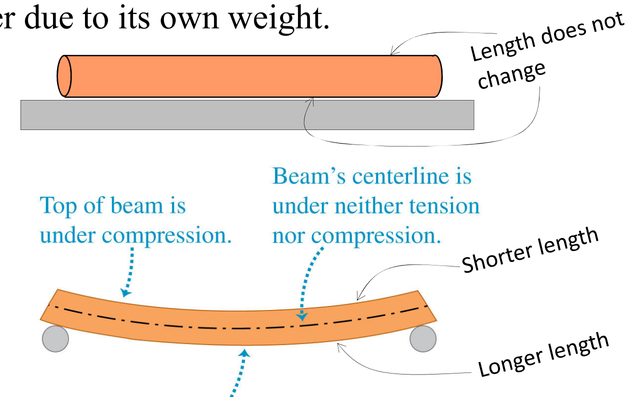
Elongation

Cross-sectional area of object

Likewise for compression

Compression and Tension

- In many situations, objects can experience both tensile stress and compressive stress at the same time.
- For example, a horizontal beam supported at both ends sags at the center due to its own weight.



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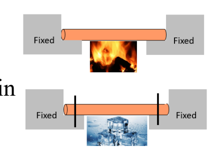
Formulation for Thermal Stress $\left(\frac{F}{A}\right)$



- Assume that a long cylinder is constrained in a fixed space (or pinned at both ends).
- The stress in the cylinder is compressive if ΔT is positive (temperature is increased in the fixed space).
- The fractional change (ratio) in length if the cylinder is free (not in the fixed space and not pinned) is

$$\Delta L = \alpha L_i \Delta T \rightarrow \left(\frac{\Delta L}{L_i}\right)_{\text{thermal}} = \alpha \Delta T. \quad \left(\frac{\Delta L}{L_i}\right)_{\text{thermal}}$$ can be positive (elongation), or negative (contraction).

- In order to maintain the same length, the fixed space (or the pins) must produce the equal and opposite fractional change in length. From the Young's modules

$$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L_i}\right)} \rightarrow \left(\frac{\Delta L}{L_i}\right)_{\text{by agents}} = \frac{F}{AY}$$

- As the length remains a constant in the constrained space (or as being pinned at both ends):

$$\left(\frac{\Delta L}{L_i}\right)_{\text{thermal}} + \left(\frac{\Delta L}{L_i}\right)_{\text{by agents}} = 0 \rightarrow \frac{\alpha \cancel{L_i} \Delta T}{\cancel{L_i}} + \frac{F}{AY} = 0 \rightarrow \alpha \Delta T + \frac{F}{AY} = 0 \rightarrow \frac{F}{A} = -Y\alpha\Delta T$$

Thermal stress:

Force needed to keep length of rod constant

$$\frac{F}{A} = -Y\alpha\,\Delta T$$

Young's modulus

Temperature change

Coefficient of linear expansion

Cross-sectional area of rod

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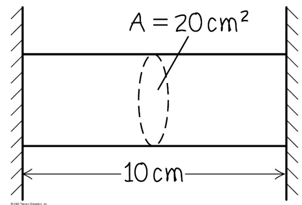
Problem. An aluminum cylinder 10 cm long, with a cross-sectional area of 20 cm², is used as a spacer between two steel walls. At 17.2°C it just slips between the walls. Calculate the thermal stress in the cylinder and the total force it exerts on each wall when it warms to 22.3°C, assuming that the walls are perfectly rigid and a constant distance apart.



$$\left(Y_{\text{aluminum}} = 7.0\times10^{10}\ \text{N/m}^2,\ \alpha_{\text{aluminum}} = 2.4\times10^{-5}\ (\text{°C})^{-1}\right)$$

Answer:

Thermal Stress $\dfrac{F}{A} = -Y\alpha\Delta T$

$$= -\left(7.0\times10^{10}\ \text{N/m}^2\right)\left(2.4\times10^{-5}\ (\text{°C})^{-1}\right)\left(22.3\text{°C} - 17.2\text{°C}\right)$$

$$= -8.57\times10^{6}\ \text{N/m}^2$$ (The aluminum cylinder will have compressive stress as it is constrained for expansion.)

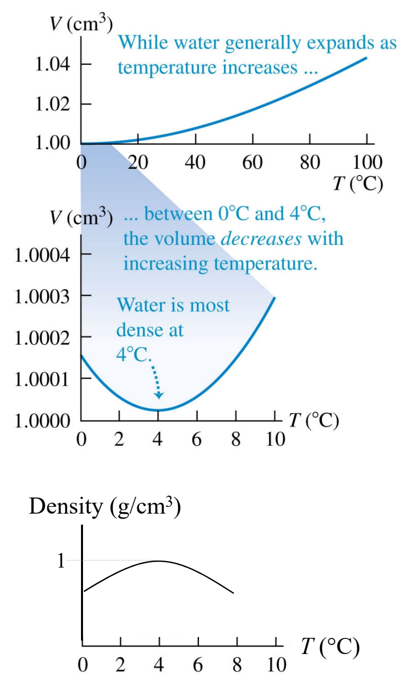
Based on the Newton's Third Law, total force exerted on each wall is

$$F = \left(\frac{F}{A}\right)_{\text{thermal}} \times A = \left(8.57\times10^{6}\ \text{N/m}^2\right)\times\left(20\times10^{-4}\ \text{m}^2\right) = 1.71\times10^{4}\ \text{N}$$

The stress on the cylinder and the force it exerts on each wall are immense.
Such thermal stresses must be accounted for in engineering!!

Thermal Expansion of Water

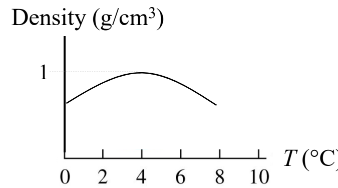
- Between 0°C and 4°C, water **decreases** in volume with increasing temperature (or between 4°C down to 0°C , water **increases** in volume (decreasing density) with decreasing temperature.)
- Because of this anomalous behavior, lakes freeze from the top down instead of from the bottom up.



Volume of Water
for the fixed Mass

Temperature

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$



Density
of Water

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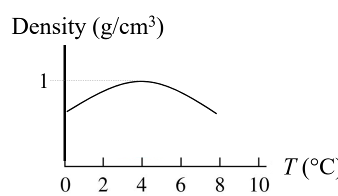
Question. How can fish survive as water freezes in the pond?

Answer:

The water in a pond begins freezing at the surface rather than the bottom. When the atmospheric temperature drops from, say 8°C to 4°C, the surface water also cools and consequently decreases in volume (as density goes up). This means that the surface water is denser than the water below it, which has not cooled and not as dense at the water on top. As a result, the surface water sinks, and the water from below is forced to the surface to be cooled. This is the circulation.



But when the atmospheric temperature decreases further from 4°C to 0°C, the surface water expands as it cools, becoming less dense than the water below it – so the mixing process (or the circulation process) stops and eventually the surface water freezes.

As the water freezes (0°C), the ice remains on the surface because ice is less dense than water. The ice continues to build up at the surface, while water near the bottom remains at 4°C so the fish can still survive.

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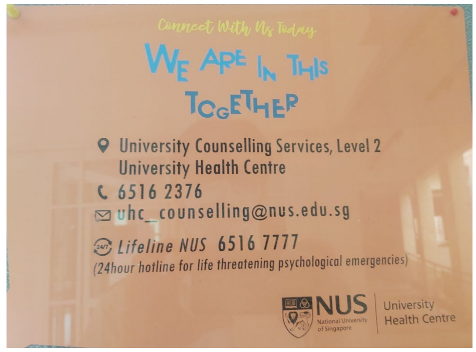
Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

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